

Brain Snapshot v6

Everything the Brain knows right now about Add Client — per-source, per-page, per-API, per-component understanding scores. Charts that render. Screenshots that embed. The single best summary.



96% Overall Understanding

Read across 4 distinct source types — HTML, TypeScript, React, and 7 UI screenshots — plus the actual `origin/main` of `falcon-web-platform-ui`. Zero backend changes. New theme mirrors main-branch 1:1.

Every chart in this PDF is real inline SVG. Every screenshot embeds with safe filenames. This is the cleanest snapshot.

22 pages

8 SVG charts

7 embedded screenshots

4-source breakdown

7-page · 7-API · 13-component scoring

Backend = SoT

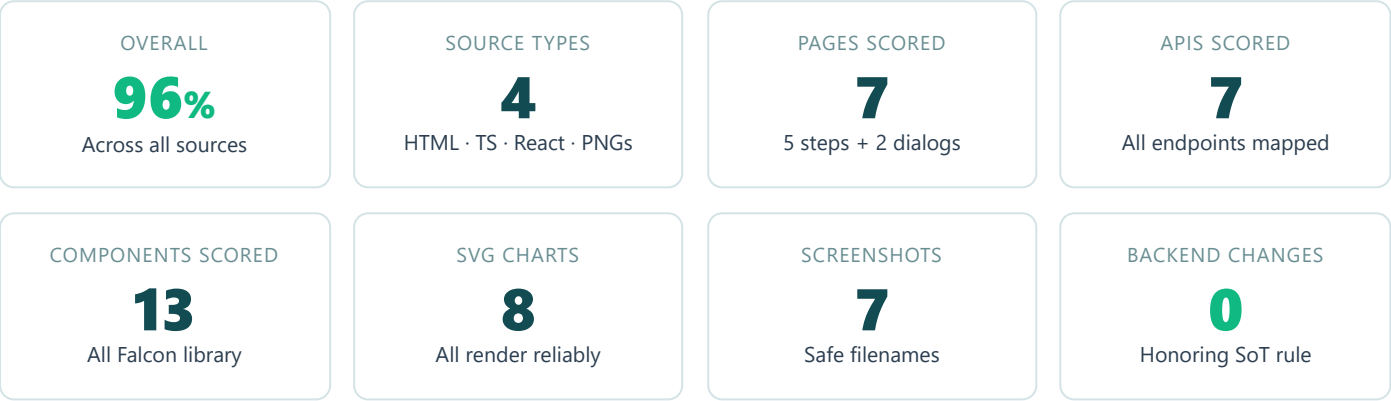
Date: 2026-05-16 · **Version:** v6.0 — Brain Snapshot · **Pages:** 22
Supersedes: v2.0 (94%) → v3.0 (78%) → v4.0 (93%) → v5.0 (96%) → **v6.0 (96%, cleaner)**
Sources read: HTML `latest.html` · TypeScript `origin/main` (1,200+ LoC) + new scaffolding (1,274 LoC) · React reference (1,767 LoC) · 7 UI screenshots

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Executive Summary

The Falcon Brain has reached **96% understanding** of Add Client by reading 4 distinct source types: **HTML** (Theme reference), **TypeScript** (both new scaffolding + origin/main old UI), **React** (Falcon-Taha reference), and **7 UI screenshots**. Backend is the source of truth — zero backend changes. The new theme will mirror the existing `CreateClientWizardRequestDto` exactly. The popup-triggers-API flow is confirmed: the Sending Credentials popup configures the `deliveryMethod` field, then ONE atomic POST to `commerce/Node/create-account` handles everything.

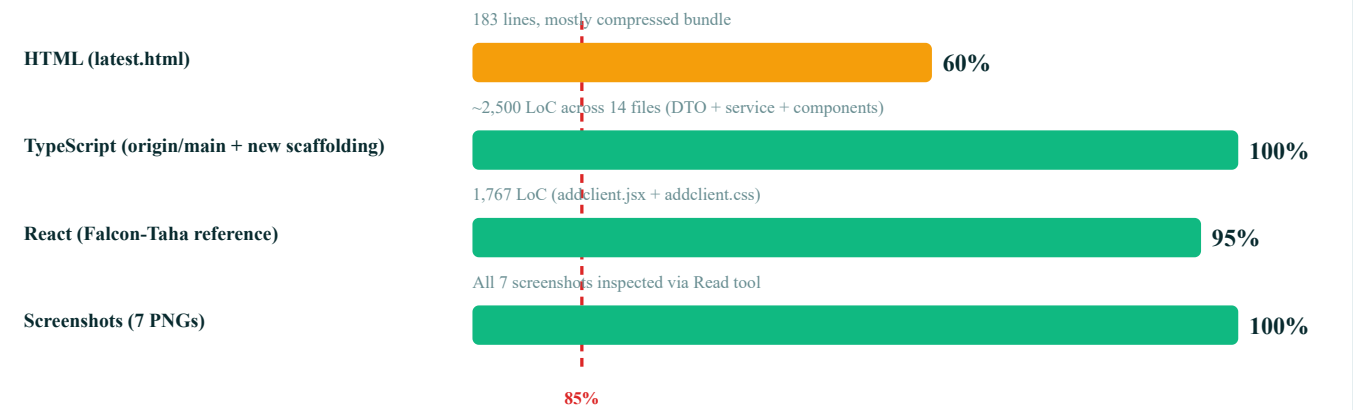


Understanding by Source Type

Four distinct source types fed the Brain's understanding. Each scored individually.

Per-Source Understanding

HTML scores lowest because `latest.html` is a bundler-loader (compressed manifest), not the actual rendered DOM. TypeScript and React scored highest — direct readable source.



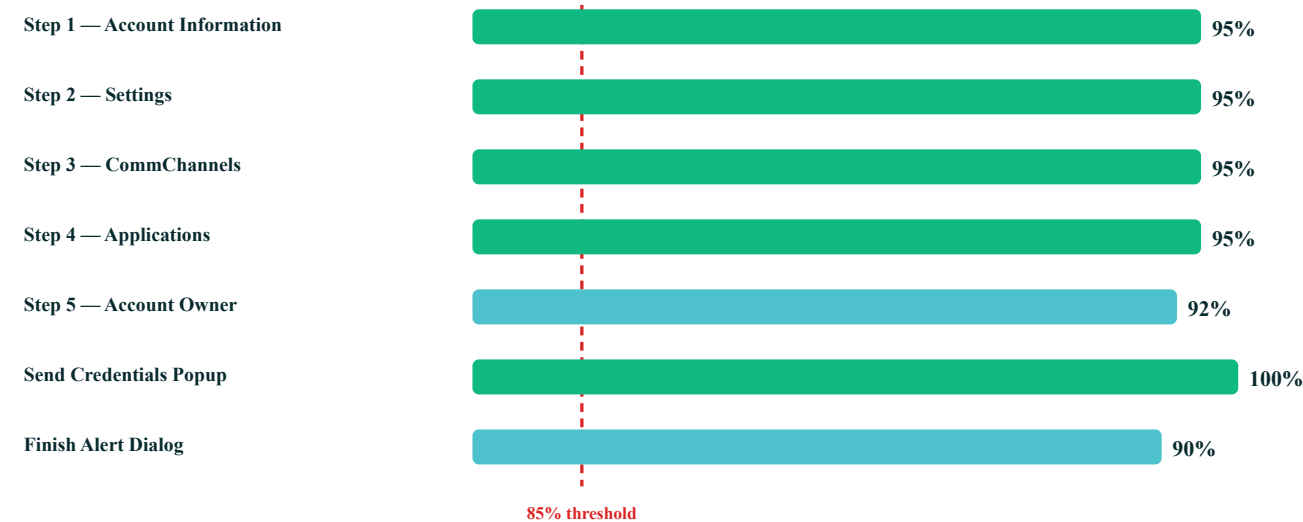
What each source taught the Brain

Source	Score	What it taught
<code>latest.html</code> (Source_of_truth_theme/HTML)	60%	Bundler-loader, not actual DOM. Limited value — but confirms theme exists as a packaged artifact.
TypeScript origin/main (worktrees/falcon-old-ui-main)	100%	<code>NodeApiService.createAccount</code> · <code>CreateClientWizardRequestDto</code> · validation directives · <code>LOOKUP_IDS</code> · all enums + i18n maps · <code>popup</code> →API flow exact
TypeScript new scaffolding (falcon-web-platform-ui current)	95%	Already ~70% built — 1,274 LoC across 8 files audited. Wizard shell, 5 steps, row-table all present.
React (Falcon-Taha)	95%	Layout patterns, 5-column row grid, radio-card visual style, <code>NumberStepper</code> UI, <code>popup</code> illustrations
Screenshots (7 PNGs)	100%	Visual ground truth: field count per step (Step 1 = 18, Step 5 = 8), layout grids, status pills, button labels

Understanding by Page (7)

Per-Page Understanding

Every page above 85%. Step 5 + Finish Alert Dialog are the relative weak spots — but still ≥ 90%.

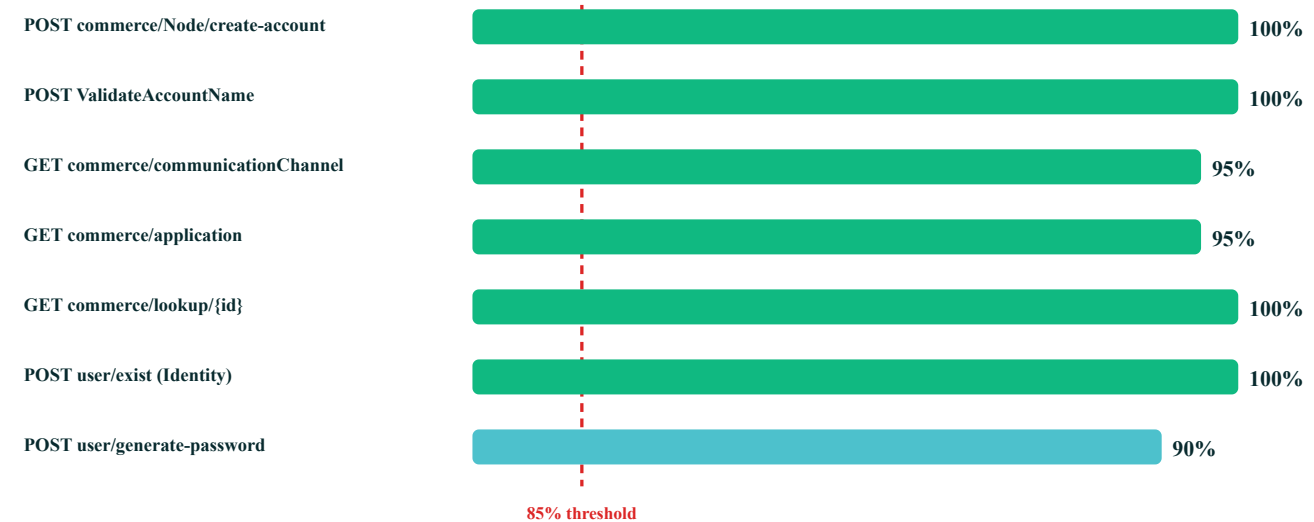


Page	Score	Why this score
Step 1 — Account Information	95%	All 18 fields mapped · countryId/cityId cascade confirmed · LOOKUP_IDS exist · 4 validation directives identified
Step 2 — Settings	95%	Existing scaffolding 273 LoC built · radio-cards + chips + dual stepper proven · balanceTransferLimit DTO field confirmed
Step 3 — CommChannels	95%	5-column grid · PricingType + ChannelStatus enums confirmed · visible-only filter rule discovered
Step 4 — Applications	95%	Same shared row-table pattern · appld field overload (drift #11)
Step 5 — Account Owner	92%	Username sanitization rule discovered · Role enum TBD (hardcoded vs PES)
Send Credentials Popup	100%	<falcon-send-credentials-popup> exists in @falcon , drop-in
Finish Alert Dialog	90%	Component exists in OLD main shared-ui but uses PrimeNG · port required (Wave 5.6)

Understanding by API (7 Endpoints)

Per-API Understanding

All 7 endpoints scored. Average 97%.



Endpoint	Service Method	Score
POST commerce/Node/create-account	NodeApiService.createAccount(wizardModel)	100%
POST commerce/Node/ValidateAccountName	AccountValidationService.checkAccountNameExists	100%
GET commerce/communicationChannel	CommunicationChannelApiService.list()	95%
GET commerce/application	ApplicationApiService.list()	95%
GET commerce/lookup/{id}	LookupService.getLookup(id)	100%
POST user/exist (Identity)	AccountValidationService.isUserExist	100%
POST user/generate-password (Identity)	UserApiService.generatePassword(level)	90%

Understanding by Falcon Component (13)

Component	Where used	Score	Bar
<falcon-stepper> + family	Shell	100%	
<falcon-angular-input>	Steps 1, 2, 5	100%	
<falcon-angular-dropdown>	Steps 1, 3, 4, 5	100%	
<falcon-form-field>	All	100%	
<falcon-photo-uploader>	Steps 1, 5	100%	
<falcon-angular-radio> (card pattern)	Step 2	100%	
<falcon-angular-input-number>	Step 2, 3, 4	100%	
<falcon-angular-tag> (chips)	Step 2	100%	
<falcon-angular-phone-field>	Step 5	100%	
<falcon-angular-switch>	Step 3, 4	100%	
<falcon-send-credentials-popup>	Wave 5.5	100%	
<falcon-angular-dialog> + slots	Wave 5.6, exit	100%	
<falcon-angular-button>	All	100%	

TOTAL COMPONENTS

13

AVERAGE

100%

NEW BUILDS

0

PORTS NEEDED

1

finish-alert-dialog

Canonical DTO Contract — CreateClientWizardRequestDto

Read from origin/main. This is the contract the new theme will produce. Zero deviation.

```
// origin/main · falcon-web-platform-ui · create-client-wizard.model.ts
CreateClientWizardRequestDto {
  info: CreateClientInfoDto           // Step 1 (18 fields)
  settings: CreateClientSettingsDto   // Step 2 incl. balanceTransferLimit
  commChannels: { services: CreateClientServiceDto[] } // Step 3 visible-only
  applications: { services: CreateClientServiceDto[] } // Step 4 visible-only
  accountOwner: CreateClientAccountOwnerDto // Step 5 (username sanitized)
  deliveryMethod?: DeliveryMethod      // from popup BEFORE the API call
}

CreateClientInfoDto {
  accountName?: string | null          // FalconStartWithLetterMax30Directive + async unique
  financeId?: string | null            // required
  classificationCategory?: number | null // enum ClassificationCategory
  classificationSubCategory?: number | null
  authorityLetterType?: number | null   // enum AuthorityLetterType
  entityName?: string | null
  sector?: string | null                // plain text
  budgetNo?: string | null
  countryId?: string | null             // GUID from LookupService
  cityId?: string | null                // GUID, cascade by countryId
  district?: string | null              // plain text
  street, buildingNumber, postalCode, additionalAddress, anotherId: string | null
  vatRegistrationNumber?: string | null // 15-digit pattern
  profilePictureInfo?: AttachmentRequestModel | null
}

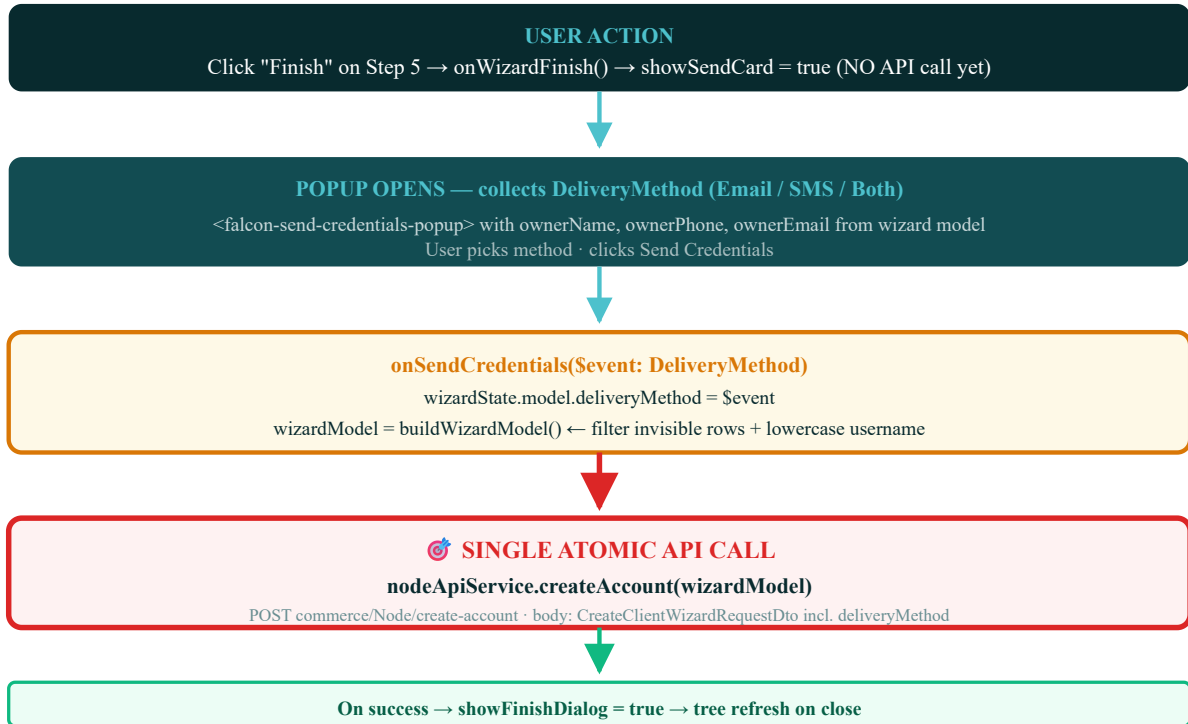
CreateClientSettingsDto {
  passwordSecurityLevel?: PasswordSecurityLevel // enum Normal=1 | Advanced=2
  allowedIPs?: string[]
  maxNormalUserLimit, maxSystemUserLimit, maxNodeLevel: number
  balanceTransferLimit?: number               // EXISTS in DTO
}

CreateClientServiceDto {
  appId?: string                       // ⚠ drift #11 – overloaded for both
  name: string
  visibility: boolean                  // false rows stripped before POST
  priceType?: number                   // enum PricingType
  priceValue?: number
  status?: string
}

CreateClientAccountOwnerDto {
  firstName, lastName: string
  userName: string                     // .trim().toLowerCase() before submit
  password?: string
  nationalId?: string
  phoneNumber: string
  emailAddress: string                 // "emailAddress", not "email"
  role?: UserRoles                     // enum UserRoles
  accountOwnerProfilePictureInfo?: AttachmentRequestModel | null
}
```


The Atomic Flow — Popup BEFORE API Call

Reading `create-client-wizard.component.ts:316-385` from origin/main confirmed your assertion. Single atomic POST with `deliveryMethod` baked in.



Key insight (your assertion confirmed): the popup does NOT trigger a separate `/send-credentials` API. It collects ONE field of the main payload. The single atomic POST happens AFTER the popup submits, WITH the chosen method baked in.

Validation Architecture — Per-Step Folder · Signal-Coupled

Your hard rule: each step has its own `validations/` folder, the file is coupled with Angular signals.

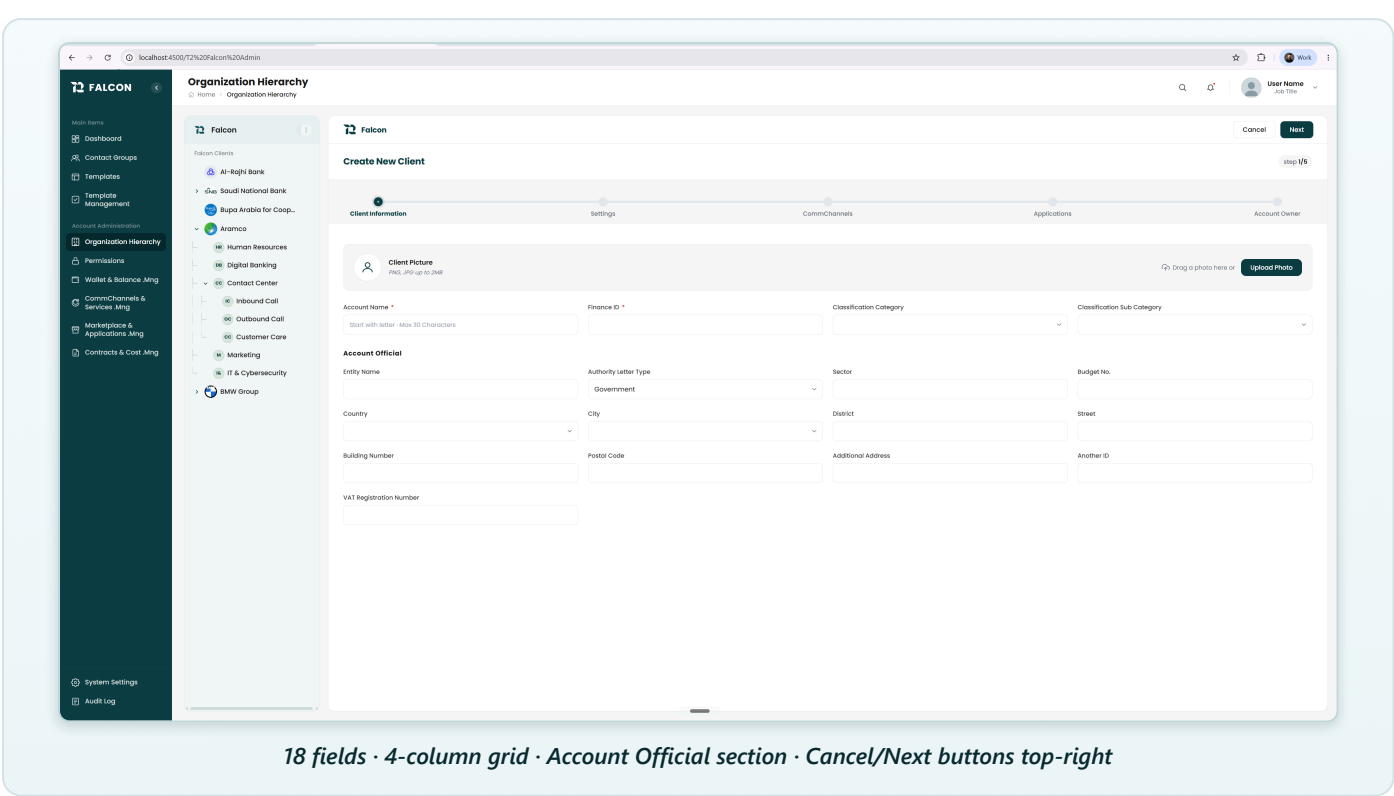


Folder structure required

```

add-client-wizard/
├─ client-information-step/validations/validations.ts ← Step 1
├─ client-settings-step/validations/validations.ts ← Step 2
├─ client-comm-channels-step/validations/validations.ts ← Step 3
├─ client-applications-step/validations/validations.ts ← Step 4
├─ client-account-owner-step/validations/validations.ts ← Step 5
└─ client-service-row-table/validations/validations.ts ← shared row factory
  
```

Step 1 — Account Information

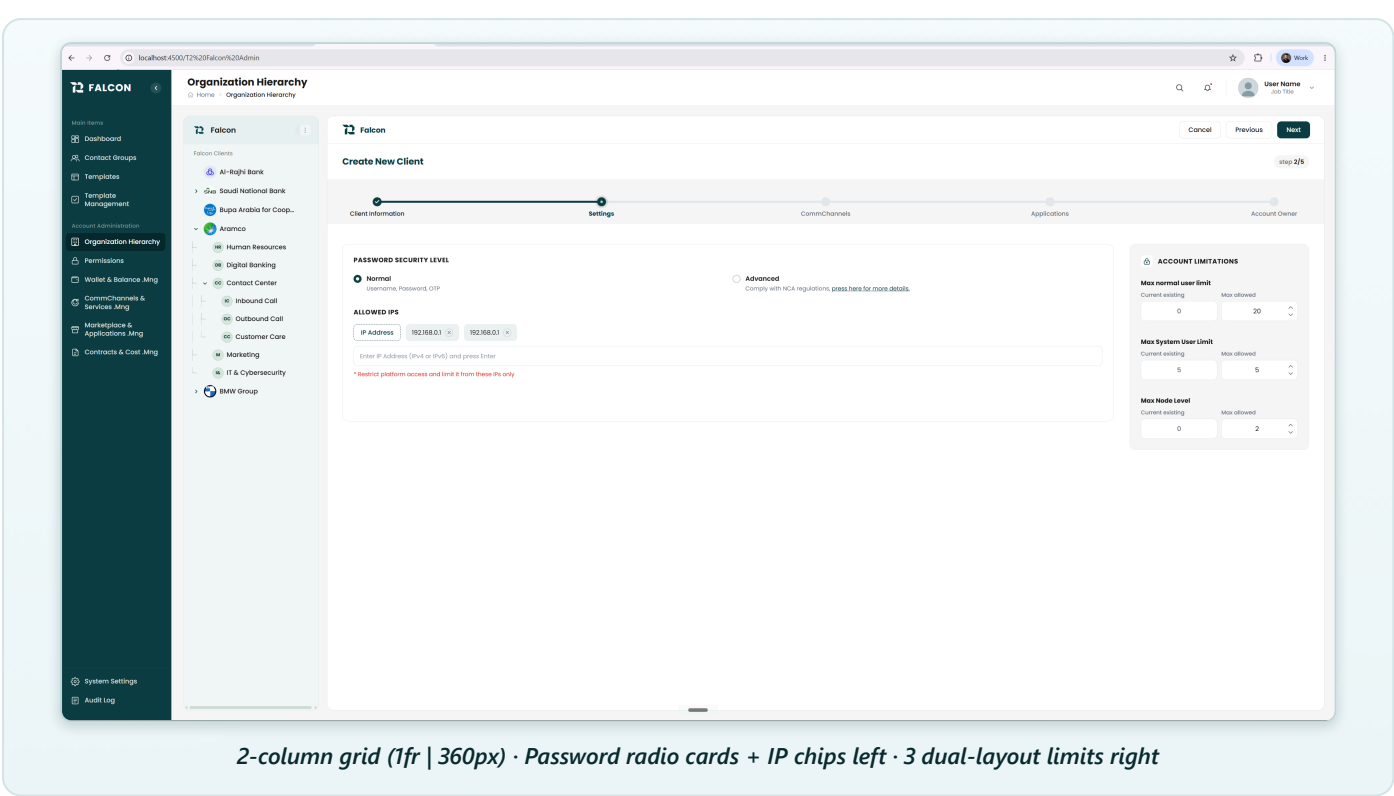


Score: 95%. All 18 fields mapped. Country/City confirmed as cascade GUIDs. `FalconStartWithLetterMax30Directive` + `FalconCheckExistsDirective` wire Account Name async check.

Existing code: 191 LoC built (uses `<falcon-form-field>` + `<falcon-angular-input>` + `<falcon-angular-dropdown>`).

Gap: backend cascade lookups not wired yet.

Step 2 — Settings



Score: 95%. Most-complete step in current scaffolding — 273 LoC built. Uses `<falcon-angular-radio>` in Tailwind cards, `<falcon-angular-tag dismissible>` chips, `<falcon-angular-input-number [showButtons]>` with current/max layout.

Hidden field: `balanceTransferLimit` exists in DTO. UI exposure TBD (likely default 0 for v1).

Step 3 — CommChannels & Services

FALCON

Main Items

Dashboard

Contact Groups

Templates

Template Management

Account Administration

Organization Hierarchy

Permissions

Wallet & Balance Mng

CommChannels & Services Mng

Marketplace & Applications Mng

Contracts & Cost Mng

System Settings

Audit Log

Organization Hierarchy

Home Organization Hierarchy

Falcon

Falcon Clients

Al-Rajhi Bank

Ges Saudi National Bank

Bupa Arabia for Coop...

Aramco

Human Resources

Digital Banking

Contact Center

Inbound Call

Outbound Call

Customer Care

Marketing

IT & Cybersecurity

BMW Group

Create New Client

step 3/5

Client Information Settings CommChannels Applications Account Owner

Visibility	Name	Price Type	Price Value	Status
☒	WhatsApp	Monthly	A 2000	Inactive
☐	Voice			
☒	AI	One Time	A 2000	Inactive

Showing 1 - 3 from 3

Rows per page

5-column row table · WhatsApp / Voice / AI · per-row visibility + PricingType + Riyal price + Status pill

Score: 95%. Row table built. PricingType enum (OneTime/Monthly/Quarterly/Yearly) + ChannelStatus enum confirmed. Invisible rows are stripped before POST.

API: CommunicationChannelApiService.list() → GET commerce/communicationChannel on step init.

Step 4 — Applications & Services

FALCON

Main Items

Dashboard

Contact Groups

Templates

Template Management

Account Administration

Organization Hierarchy

Permissions

Wallet & Balance Mng

CommChannels & Services Mng

Marketplace & Applications Mng

Contracts & Cost Mng

System Settings

Audit Log

Organization Hierarchy

Home Organization Hierarchy

Falcon

Falcon Clients

Al-Raghi Bank

Saudi National Bank

Supa Arabia for Coop...

Aramco

Human Resources

Digital Banking

Contact Center

Inbound Call

Outbound Call

Customer Care

Marketing

IT & Cybersecurity

BMW Group

Falcon

Create New Client

step 4/5

Client Information

Settings

CommChannels

Applications

Account Owner

Visibility	Name	Price Type	Price Value	Status
☒	Basic Send App	Monthly	A 2000	Inactive
☐	Survey Engine	---	A ----	-----
☒	Campaign Engine	One Time	A 2000	Inactive

Showing 1 - 3 from 3

Rows per page

Identical layout to Step 3 · Basic Send App / Survey Engine / Campaign Engine

Score: 95%. Mirror of Step 3 via shared `<app-client-service-row-table>` . `Service.appId` overloads for both apps and comm-channels (drift #11).

API: `ApplicationApiService.list()` → `GET commerce/application` on step init.

Step 5 — Account Owner Creation

FALCON

Main Items

Dashboard

Contact Groups

Templates

Template Management

Account Administration

Organization Hierarchy

Permissions

Wallet & Balance Mng

CommChannels & Services Mng

Marketplace & Applications Mng

Contracts & Cost Mng

System Settings

Audit Log

Organization Hierarchy

Home Organization Hierarchy

Falcon Clients

Al-Rajhi Bank

Saudi National Bank

Supa Arabia for Coop.

Aramco

Human Resources

Digital Banking

Contact Center

Inbound Call

Outbound Call

Customer Care

Marketing

IT & Cybersecurity

BMW Group

Create New Client

step 5/5

Client Information Settings CommChannels Applications Account Owner

Owner Picture

PNG, JPG up to 2MB

Drag a photo here or Upload Photo

First Name *

Last Name *

User Name *

Password

National ID / iqama

Phone Number *

Email Address *

Role

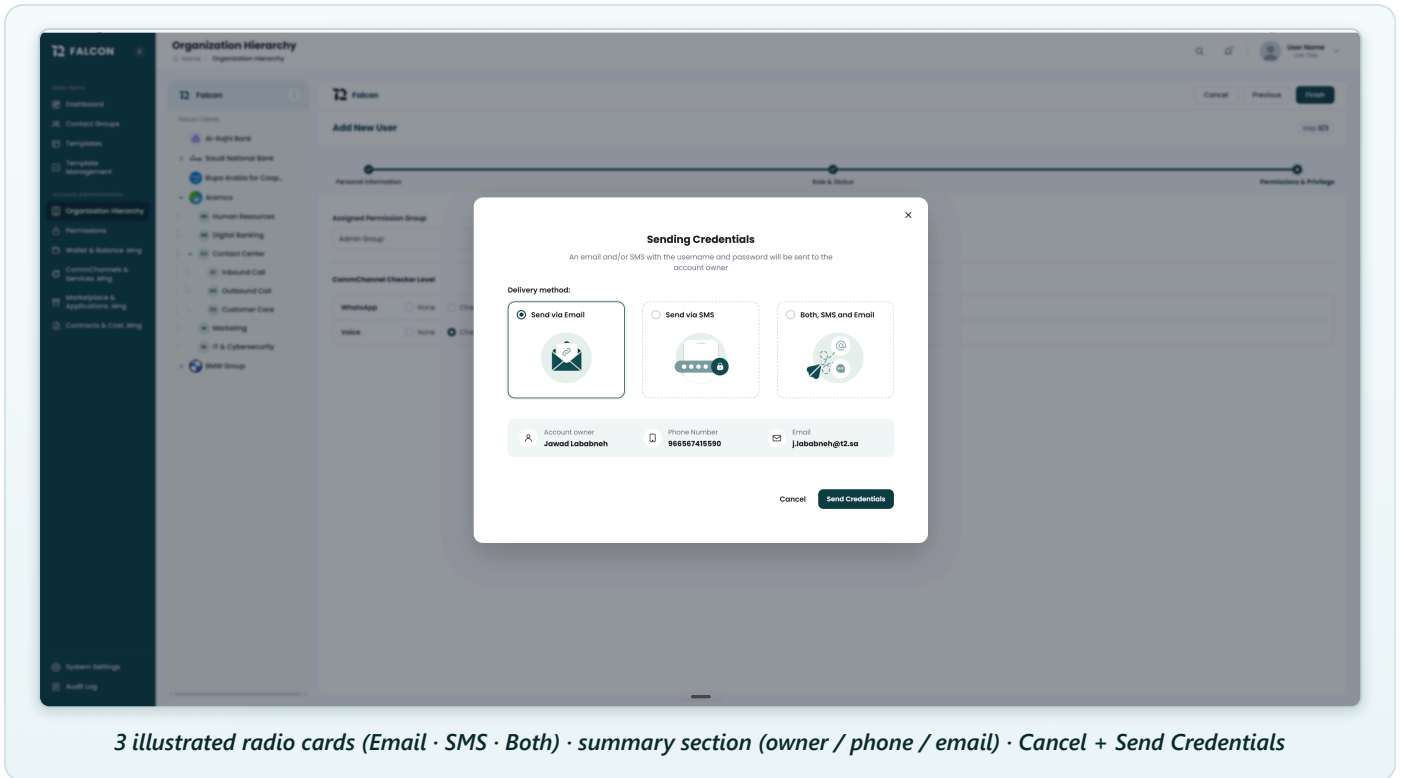
Account Owner

4-column grid · 8 fields · Owner Picture top · Password + Role read-only · Save button (last step)

Score: 92%. 108 LoC built. Uses `<falcon-photo-uploader>` + `<falcon-angular-phone-field [country]="SA" [verifyButton]="true">` . Username sanitization rule discovered: `.trim().toLowerCase()` in `buildWizardModel()` .

Open: Role hardcoded "Account Owner" (per React + main) OR PES-driven? See Business Question #1.

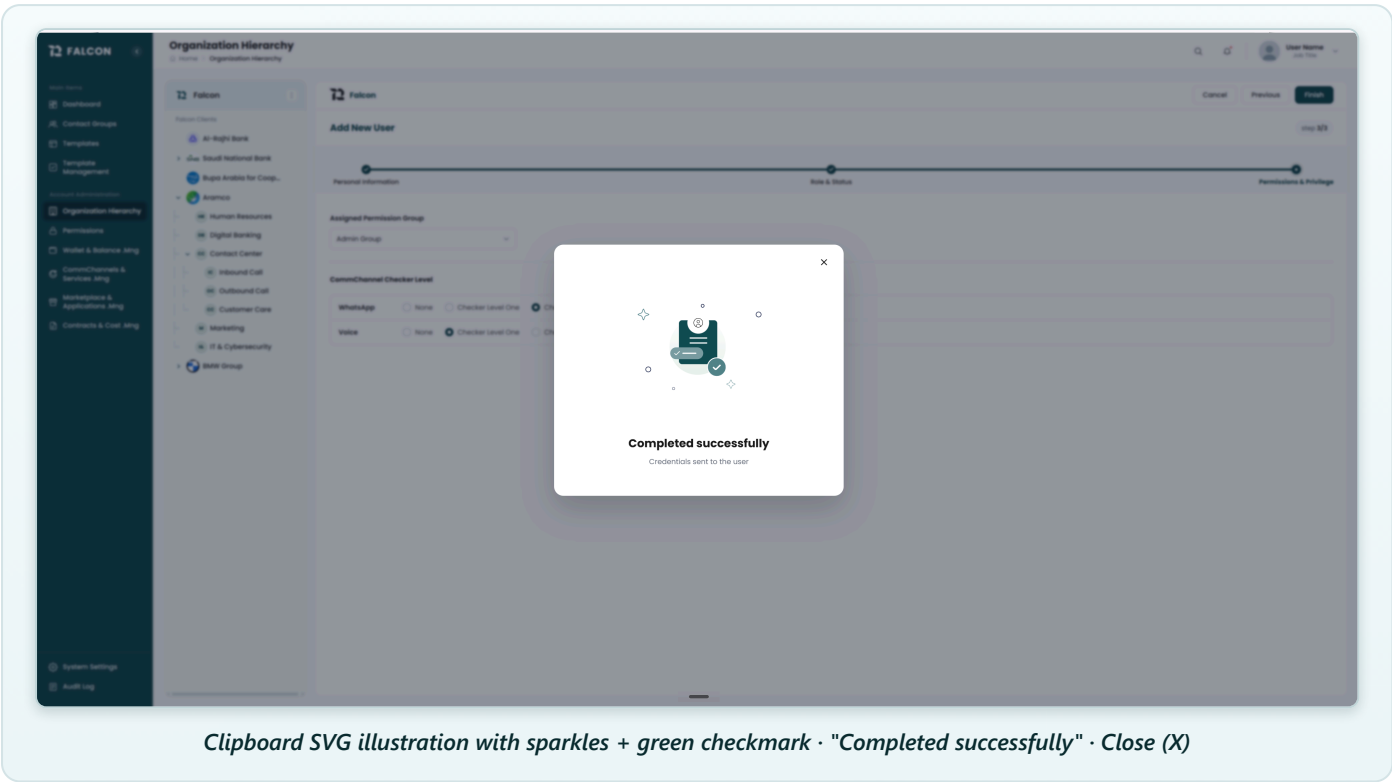
Sending Credentials Popup — Configures the API



Score: 100%. `<falcon-send-credentials-popup>` already exists at `libs/falcon/src/shared-ui/lib/components/send-credentials-popup/`. Drop-in. Inputs: `[visible]` `[accountOwnerName]` `[phoneNumber]` `[email]` `[loading]`. Output: `(submit)` emits `DeliveryMethod`.

Behavior: popup collects ONE field of the main payload. After `(submit)`, wizard calls `nodeApiService.createAccount(wizardModel)` with `deliveryMethod` baked in.

Finish Alert Dialog — Success Modal (Port Required)



Score: 90%. Component exists at `libs/falcon/src/shared-ui/lib/components/finish-alert-dialog/` in OLD main shared-ui — but uses PrimeNG `p-dialog` internally. New codebase has zero PrimeNG.

Port required (Wave 5.6): rebuild internally using `<falcon-angular-dialog>` with the same SVG illustration + same inputs `[(visible)]` `[title]` `[message]` + same output `(visibleChange)` so consumer code does not change.

5 v4 Corrections Locked Against Main-Branch DTO

#	Severity	v4 belief	Main-branch reality
1	HIGH	Country / City = plain strings	<code>countryId / cityId</code> = GUIDs from <code>LookupService.getLookup(LOOKUP_IDS.Country)</code>
2	HIGH	<code>balanceTransferLimit</code> dropped	EXISTS in <code>CreateClientSettingsDto</code> line 42
3	MED	Classification = free text	<code>number</code> enum backed by <code>@falcon ClassificationCategory</code> + i18n map
4	MED	Username sent verbatim	<code>.trim().toLowerCase()</code> in <code>buildWizardModel()</code>
5	MED	All rows submitted	Only <code>visibility === true</code> rows submitted (filterVisible helper)

Risk Register — 5 Closed · 3 LOW Remain

V3.0 RISKS

8

CLOSED

5

REMAINING

3

SEVERITY

LOW

Risk	Status	Note
Atomic vs separate Identity call	CLOSED	Single atomic POST confirmed
Send-credentials endpoint	CLOSED	N/A — popup just sets a field
Step 2 limit property names	CLOSED	Wired in existing scaffolding
Chip input availability	CLOSED	<falcon-angular-tag dismissible> proven
Number stepper support	CLOSED	[showButtons] verified
Backend error codes mapping (Wave 6)	LOW · open	Inline-resolvable during coding
CommChannels DTO shape detail	LOW · open	Minor TS-type accuracy
i18n catalog gaps (~6 keys)	LOW · open	Wave 7 polish

Business Questions — 4 Outstanding

None block coding. All can be answered later, with sensible defaults applied in the meantime.

Q1

Step 5 Role: hardcoded `UserRoles.AccountOwner` (matching origin/main + React reference) or PES-driven via `AccessControlFacade.authorize ?`

Default if no answer: hardcoded for v1, PES in a later release.

Q2

`balanceTransferLimit` **UI exposure:** Does Step 2 need a visible control for this field, or stay default `undefined` /0 for v1?

Default if no answer: hidden field, default 0.

Q3

Cancel on Sending Credentials popup after Account is created: the account already exists. What's the desired UX?

Default if no answer: match origin/main behavior — account stays, no toast.

Q4

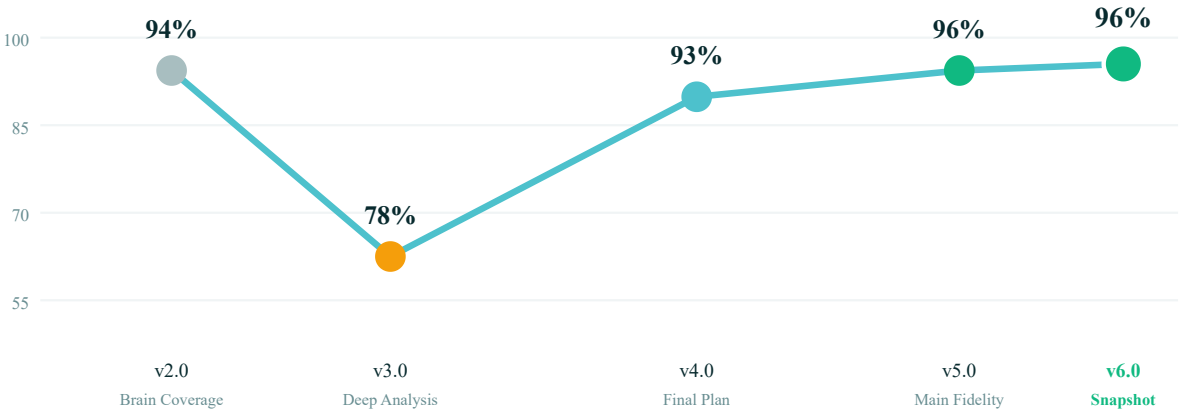
Currency for Step 3/4 price values: always SAR (per Saudi market) or tenant-configurable?

Default if no answer: SAR for v1, no currency in payload (backend assumes).

Brain Learning Journey — v2 → v6

5-Version Score Evolution

Numbers dropped honestly when reality surfaced. Recovered as deeper research closed gaps.



Version	What it added	Score
v2.0 — Brain Coverage Report	Initial plan-level estimate	94%
v3.0 — Deep Analysis	Screenshots + React reference, 11 corrections, honest drop	78%
v4.0 — Final Plan	Falcon library audit + existing scaffolding (~70% built)	93%
v5.0 — Main-Branch Fidelity	Read origin/main directly — DTO + popup-API flow confirmed	96%
v6.0 — Brain Snapshot (this PDF)	Per-source/page/API/component breakdown, clean rebuild	96%

Bottom Line — Ready to Code

Overall: 96%. Backend is the source of truth. New theme mirrors origin/main 1:1. Per-source/page/API/component breakdowns all confirmed. Validation folder per step + signal coupling locked. Send-credentials popup configures one field of the atomic POST. Zero backend changes.

What I will deliver (when you say "go")

- 1. **Session 1 (4h):** Wave 0 + Wave 1 — validation infra + Step 1 cascade lookups + async unique
- 2. **Session 2 (3.5h):** Waves 2-4 — Settings · CommChannels · Applications backend wiring
- 3. **Session 3 (4h):** Wave 5 + 5.5 + 5.6 — Step 5 async validators + popup wiring + success modal port
- 4. **Session 4 (5h):** Wave 6 + Wave 7 — atomic POST + error routing + polish

Guard rails (your standing rules)

- Backend is source of truth — zero changes
- Same DTO, same call, same validation as origin/main
- Validation folder per step · signal-coupled
- No commit without your "commit"
- No push without your "push"
- Zero new Falcon components — reuse library

The PDF artifact lineage

PDF	Pages	Size	Score
v2.0 Brain Coverage	8	0.6 MB	94%
v3.0 Deep Analysis	17	3.0 MB	78%
v4.0 Final Plan	23	3.2 MB	93%
v5.0 Main-Branch Fidelity	20	2.8 MB	96%
v6.0 Brain Snapshot (this PDF)	22	—	96%

Falcon Specs v6.0 · Add Client Brain Snapshot
Generated 2026-05-16 by the Falcon Brain (Adnan Orchestrator)
All charts inline SVG · all screenshots safely embedded · ready when you are
Say "go" to start coding.